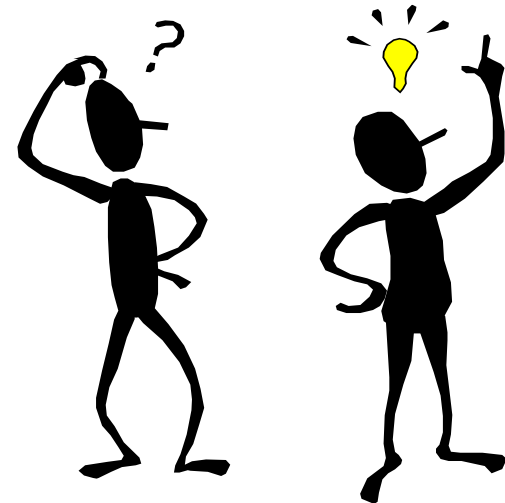


Chapter 2 Multimedia Information Representation

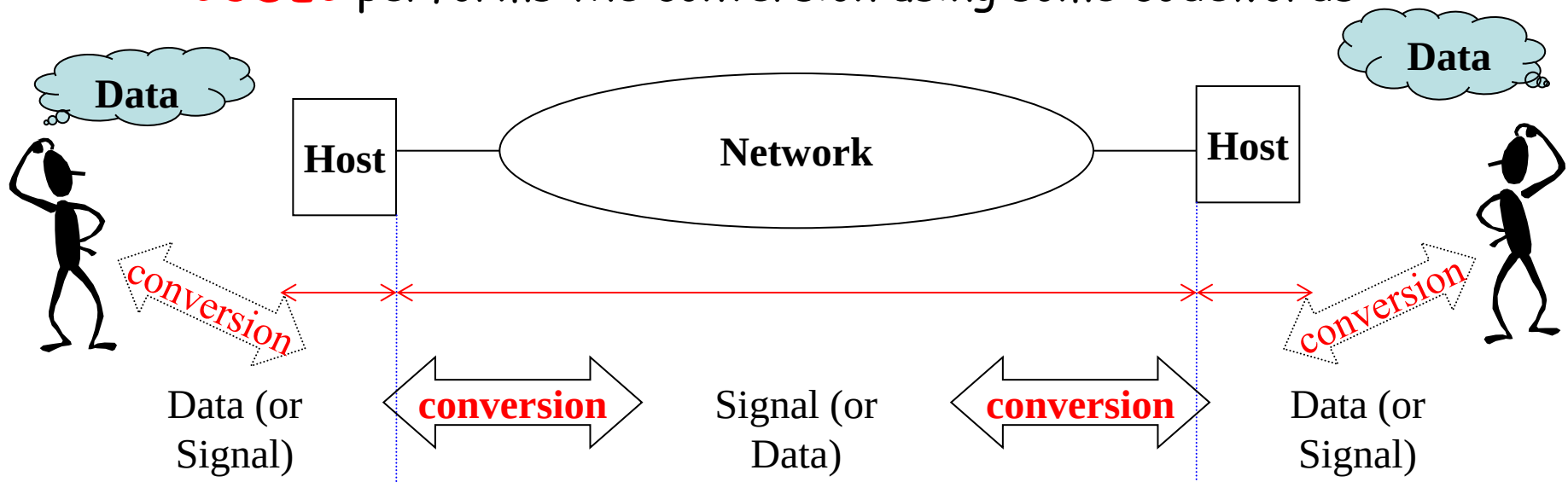
Contents

- 2.1 Introduction
- 2.2 Digitization Principles
- 2.3 Text
- 2.4 Images
- 2.5 Audio
- 2.6 Video



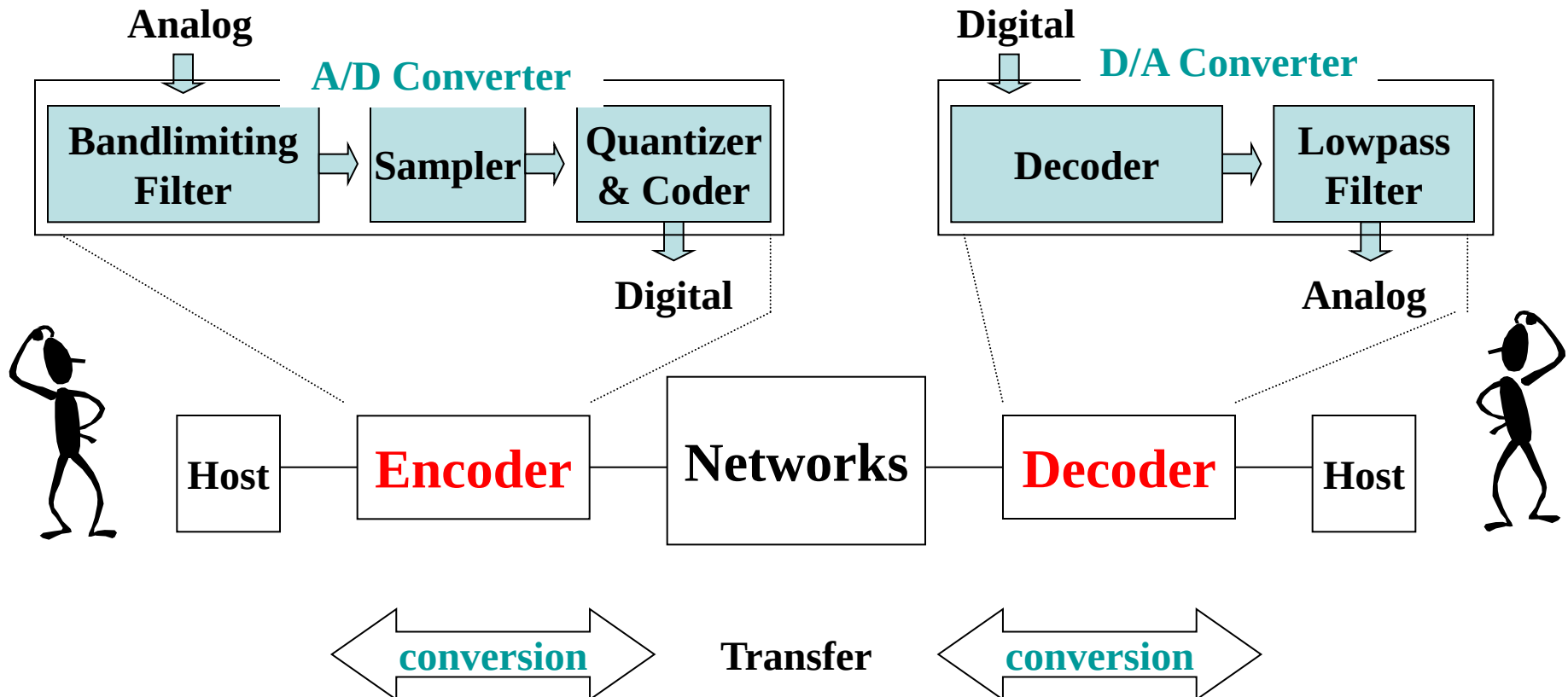
2.1 Introduction

- **Codeword**: a fixed number of bits representing a set of symbols, e.g) ASCII Code, FAX Run-length Code,
- **Signal Encoder**
- **Signal Decoder**
- **CODEC** performs the conversion using some codewords



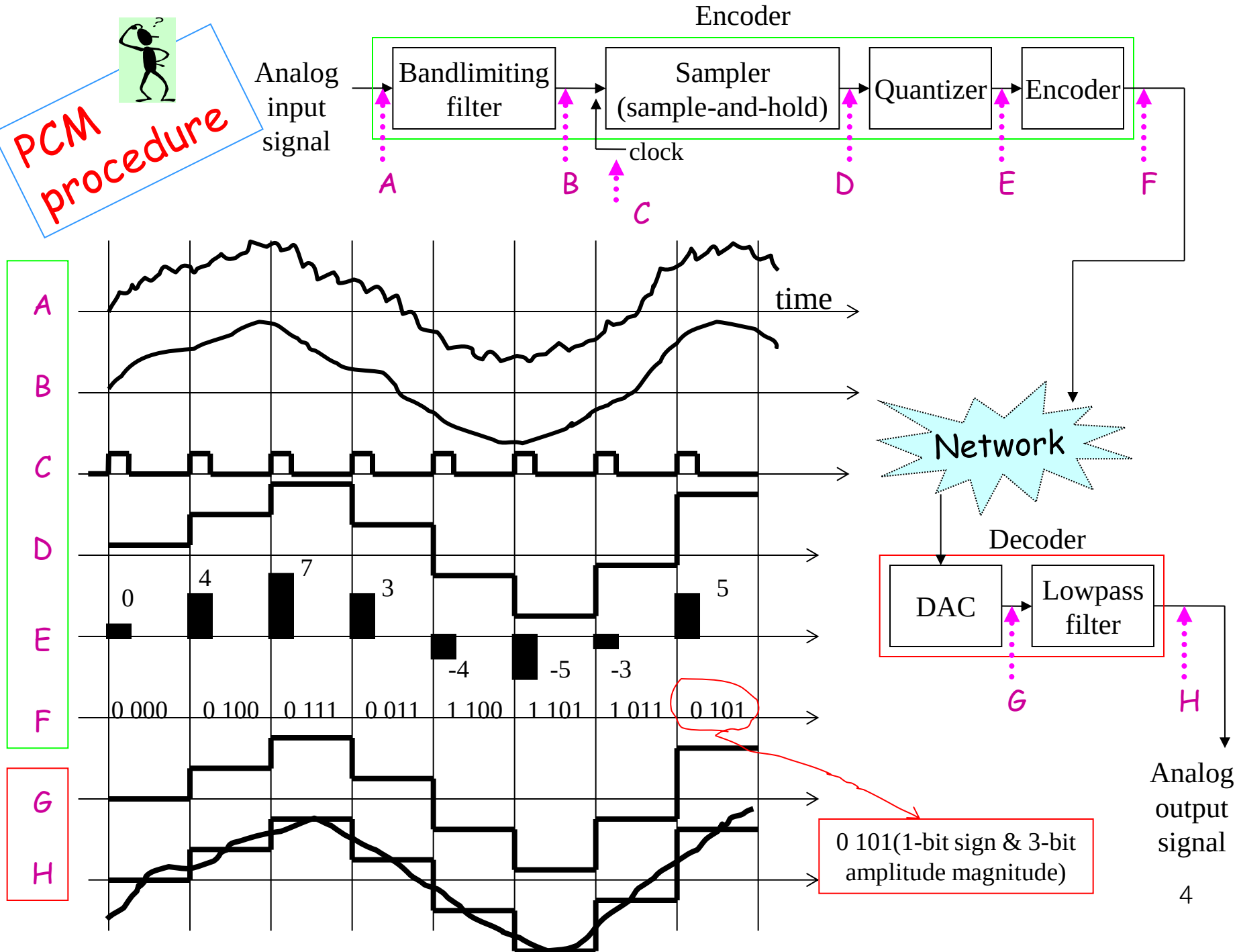
2.2 Digitization Principles (1) (Analog $\Leftrightarrow$ Digital)

- terms**
- Spectrum VS. Bandwidth
 - Signal bandwidth VS. Channel (Bandlimiting) bandwidth
 - Cutoff frequency = $\min \{ \text{Signal bandwidth, Bandlimiting bandwidth} \}$





PCM procedure



2.2 Digitization Principles(2) (Analog $\Leftrightarrow$ Digital)

- Analog Signal

- Bandwidth, B Hz, via band-limiting channel

- Encoder

- Band limiting filter
- Sampling: $2B$ sps(samples per sec) $\Rightarrow$ aliasing may happen !
- Quantizing: Aliasing filter for eliminating alias signals
 - quantization interval $q = 2(V_{max}/2^n)$
 - quantization error/noise = $\pm q/2$

Dynamic range of signal D
 $D = 20 \log_{10}(V_{max}/V_{min})$

n : # of bits

V_{max} : max(min) positive
(negative) signal
amplitude

- Decoder

- low-pass filter (= band-limiting filter = anti-aliasing filter)

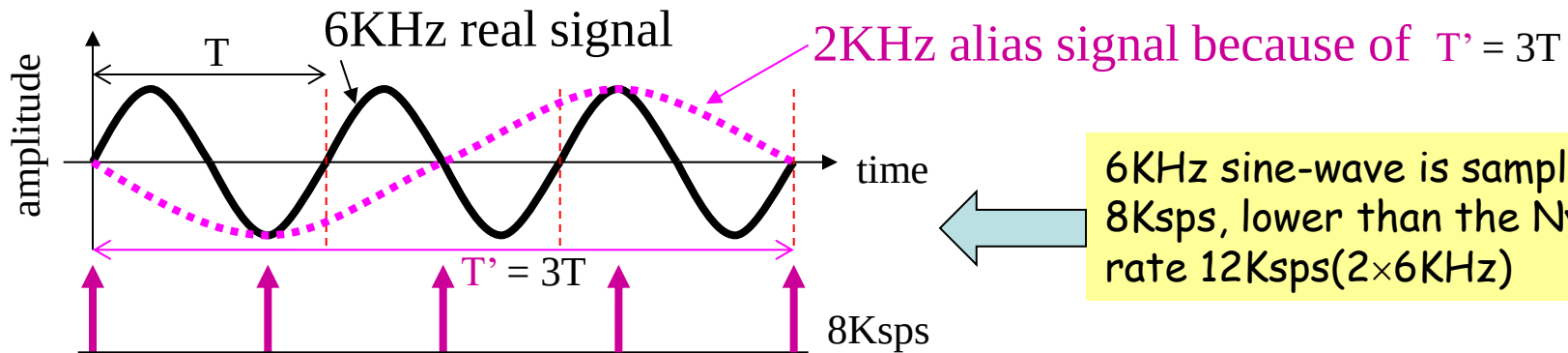
2.2 Digitization Principles (3)

(Analog $\Leftrightarrow$ Digital)

Aliasing signal & its elimination

When does aliasing occur ?

If the sampling rate is lower than the Nyquist rate



Conclusion

All frequency components in the source signal that are **higher in frequency than half the sampling frequency being used** will generate related lower-frequency alias signal which will simply add to those making up the original thereby causing it to **become distorted**

Resolution

Using “bandlimiting filter”, let’s **pass only those Frequency components up to that determined by the Nyquist rate**

bandlimiting filter = anti-aliasing filter = low-pass filter = reconstruction filter

2.2 Digitization Principles (4)

(Analog $\Leftrightarrow$ Digital)

■ Example 1

An analog signal has a dynamic range of 40 dB. Find the magnitude of the quantization noise relative to the minimum signal amplitude if the quantizer uses 1) 6 bits and 2) 10 bits

■ Solution

□ It follows that $40 = 20 \log_{10}(V_{\max}/V_{\min})$ by assumption and finally the equation $10^2 = V_{\max}/V_{\min}$ results in

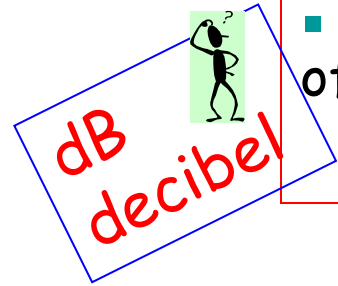
$$V_{\min} = V_{\max}/100$$

□ And the quantization noise is determined by $\pm q/2$ where, q is the quantization interval given by $q = 2(V_{\max}/2^n)$.

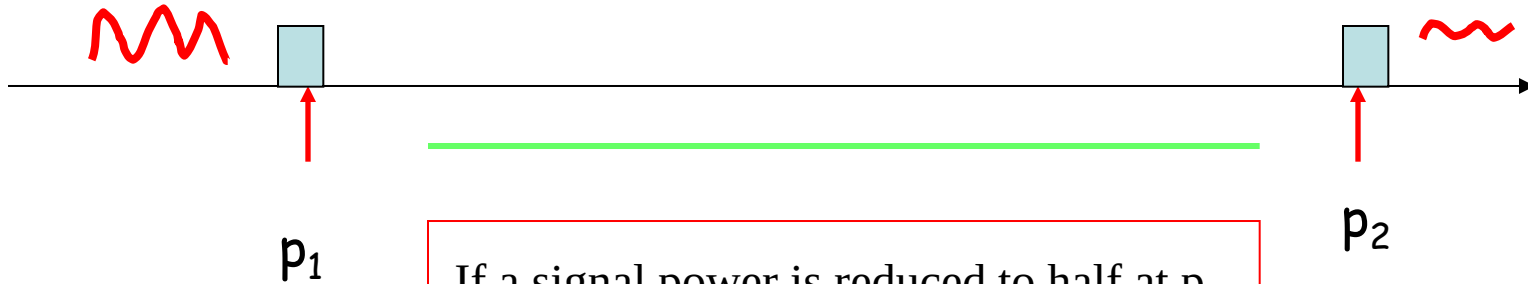
Thus $\pm q/2 = \pm V_{\max}/2^n$.

□ For $n=6$, $q/2 = \pm V_{\max}/2^6 (= \pm V_{\max}/64) > V_{\min} (= V_{\max}/100)$
→ unacceptable !

For $n=10$, $q/2 = \pm V_{\max}/2^{10} (= \pm V_{\max}/1024) < V_{\min} (= V_{\max}/100)$
→ acceptable !



■ dB (decibel) : The decibel measures the relative strength of two signals or a signal at two different points p_1 and p_2 given by $\text{dB} = 10 \log_{10}(p_2/p_1)$



If a signal power is reduced to half at p_2
such that $p_2 = p_1/2$
 $10 \log_{10}(p_2/p_1) = 10 \log_{10}(0.5p_1/p_1) =$
 $10 \log_{10}(1/2) = 10 \log_{10}1 - 10 \log_{10}2 = -3\text{dB}$

2.3 Text

- **Unformatted Text, Plaintext**

- String of fixed-size characters
- ASCII, Mosaic Characters,



*Well-defined code-words
are used for Text Creation
& Manipulation*

- **Formatted Text**

- String of characters of different sizes, styles & shapes with table, figures (graphics) & images
- Latex, Acrobat,

- **Hypertext**

- Integrated set of documents comprising formatted & unformatted texts with linkages among them
- HTML, Postscripts,

2.4 Images

- Image (still picture) Classification

- Computer-generated images (computer graphics)
e.g) palette files
- Digitized images of documents and/or pictures
e.g) fax-scanned files, scanned color-image files

VGA 640 × 480
pixels
8-bits/pixel

- Graphics

- high-level language form: description of attributes of objects
- bit-map form: actual pixel-images (VGA- Video Graphics Array)
 - gif: graphical interchange format
 - tiff: tagged image file format
 - srpg: simple raster graphics package

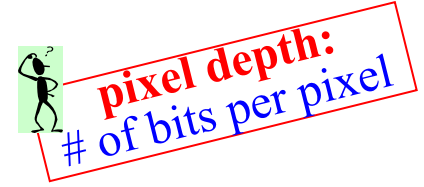


pixel (or pel):
picture element

- Digitized Documents

- Facsimile (FAX) machine, about 2Mbits/page(black-white/pixel)
 - Pixel resolution: 8 pels per mm
 - Line resolution: 3.85 or 7.7 lines per mm(100 or 200 lines per inch)

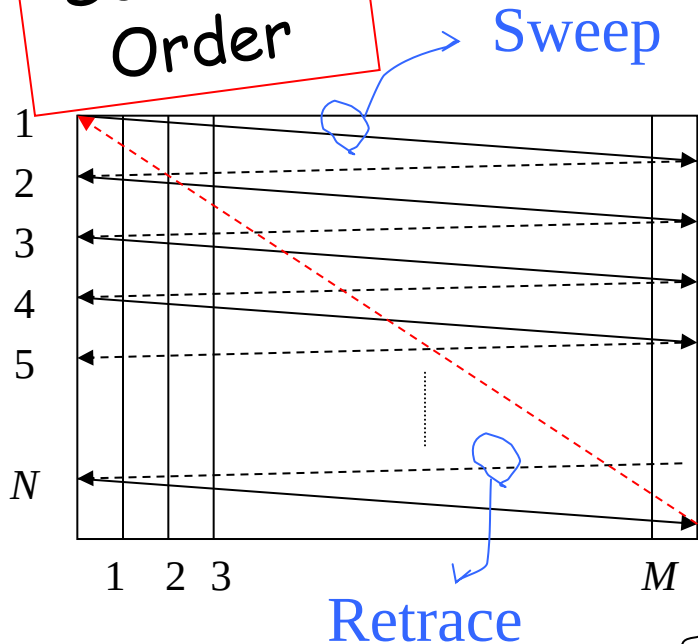
Digitized Pictures(1)



- **m-bit per pixel (pixel depth m)**
 - good-quality black-white picture: 8-bit/pixel(256 gray levels)
 - colored-picture: 24-bit/pixel(R/G/B each 8-bit yielding 16 M colors)
- **Coloring Principles : How is color produced and represented ?**
 - **Color gamut**: a whole spectrum of colors
 - **Three primary colors**: **R** (Red), **G** (Green), **B** (Blue)
 - all kind colors are produced by using different proportions of these primary colors
 - **Additive Color Mixing** on a black surface
 - **Subtractive Color Mixing** on a white surface
- **Raster-Scan Principles: TV Screen or Computer CRT Monitor**
 - **NTSC (National Television Standards Committee)-USA**
 - 525(active 480) lines/frame & 60-time refresh rate/sec
 - **PAL (Phase Alternation Line)/CCIR/SECAM**
 - 625(active 576) lines/frame & 50-time refresh/sec

Digitized Pictures(2)

Scanning Order



1. $N=525$ (National Television System Committee) & 625(Phase Alternating Line/Sequential Color with Memory/Commander's Critical Information Requirements)
2. fresh rate (Hz) = 60(NTSC) & 50(PAL/SECAM/CCIR)
3. M is determined by the aspect ratio (see the next slides)

frame : a complete set of N horizontal scan lines

frame refresh rate: # of frames per sec
at least 50 Hz to avoid flickering

Scanning Method

$M \times N$

Progressive scanning : $1 \rightarrow 2 \rightarrow 3 \rightarrow \dots \rightarrow N$: one frame

60 or 50 Hz
refresh rate

Interlaced scanning : $1 \rightarrow 3 \rightarrow 5 \rightarrow \dots \rightarrow N-1$: first half frame (field)

30 or 25 Hz
refresh rate

$2 \rightarrow 4 \rightarrow 6 \rightarrow \dots \rightarrow N$: 2nd half frame (field)

Digitized Pictures(3)



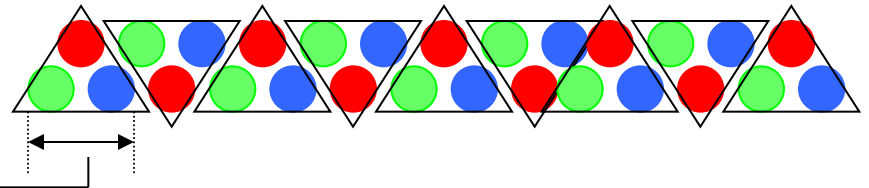
in HTML



■ Raster-Scan Principles

- **Raster**: a finely-focused electro beam
- **Phosphor**: a light-sensitive material that emits light when energized
 - **white-sensitive** phosphor: a single electron beam used
 - **color-sensitive phosphor**: each pixel comprises a set of **three** color-sensitive phosphors, one each for **R**, **G**, **B** signals, called **phosphor triad**

spot size:
0.635mm(0.025inch)



- **beam signal** may be either **analog** or **digital** form
- **Pixel Depth**: # of bits per pixel
 - **CLUT (Color Look-Up Table)**: 24-bit/pixel yields 2^{24} colors. But eye discriminates between some ranges of colors hence, each pixel value is used as an **index on CLTT** of 256 colors (**compression** achieved !)

Digitized Pictures(4)

- **Aspect Ratio**: ratio of the screen width to the screen height
 - NTSC, 525 scan lines/frame $\Rightarrow$ 480(45) data (control) lines
 - 4/3 aspect ratio $\Rightarrow 480 \times 4/3(=640)$ pixels/line
 - 16/9 aspect ratio $\Rightarrow 480 \times 16/9(=853.33)$ pixels/line
 - PAL/CCIR/SECAM 625 lines/frame $\Rightarrow$ 576(49) data (control) lines
 - 4/3 aspect ratio $\Rightarrow 576 \times 4/3(=768)$ pixels/line
 - 16/9 aspect ratio $\Rightarrow 576 \times 16/9(=1024)$ pixels/line

Representing
an $M \times N$ pixels
under a
particular
aspect ratio



Computer Graphics Array

standard	resolution	#of colors	Bytes/frame
Video GA	640 x 480 x 8	256	307.2K
ExtendedGA	640 x 480 x 16	64K	614.4K
	1024 x 768 x 8	256	786.432K
Super videoGA	800 x 600 x 16	64K	960K
	1024 x 768 x 8	256	786.432K
	1024 x 768 x 24	16M	2359.296K

refresh rate:
50-70Hz

Digitized Pictures(5)

■ Example 2.3

Derive the time to transmit the following digitized images at both 64Kbps and 1.5Mbps networks

- a $640 \times 480 \times 8$ VGA-compatible image
 - a $1024 \times 768 \times 24$ SVGA-compatible image
-

■ Solution

The size of each image in bit is as follows

- a VGA image = $640 \times 480 \times 8 = 2.46\text{Mbits}$
- an SVGA image = $1024 \times 768 \times 24 = 18.88\text{Mbits}$

The time to transmit each image is given as follows

- at 64Kbps : $VGA = 2.46\text{Mbits}/64\text{Kbps} = [2.46 \times 10^6]/[64 \times 10^3] = 38.4 \text{ sec.}$
 $SVGA = [18.88 \times 10^6]/[64 \times 10^3] = 295 \text{ sec.}$
- at 1.5Mbps: $VGA = 2.46\text{Mbits}/1.5\text{Mbps} = [2.46 \times 10^6]/[1.5 \times 10^6] = 1.64 \text{ sec.}$
 $SVGA = [18.88 \times 10^6]/[1.5 \times 10^6] = 12.59 \text{ sec.}$

Digitized Pictures(6)

■ Digital Cameras & Scanners

- (Still image cameras) 2-D grid of photo-sites (diode), light-sensitive cells, made of charge-coupled devices (CCD's)
 - level of light intensity on each photosites is converted into a digital value using an AD converter when the shutter is activated
- (Scanners) single-row of photo-sites is exposed in time-sequence with the scanning operation
- How are color images obtained ?
 - each photosite/pixel is coated with R/B/G filter & the color is determined by the level of it together with 8 neighbors in a 3 x 3 grid structure
 - General consumer
 - use of three separate exposures of a single photosite, say, first R filter, 2nd G filter, and finally B filter
 - Photo studio
 - use of three separate image sensors per pixel professional

e.g) TIFF (tagged image file format), TIFF/EP for electronic photography

2.5 Audio

■ Typical Audio Types

- **Speech signal** for interpersonal application such as (video) telephony
- **Music-quality audio** such as CD-on-demand & broadcast TV
 - synthesizer
 - microphone
 - loudspeaker

Basics on Audio Signals

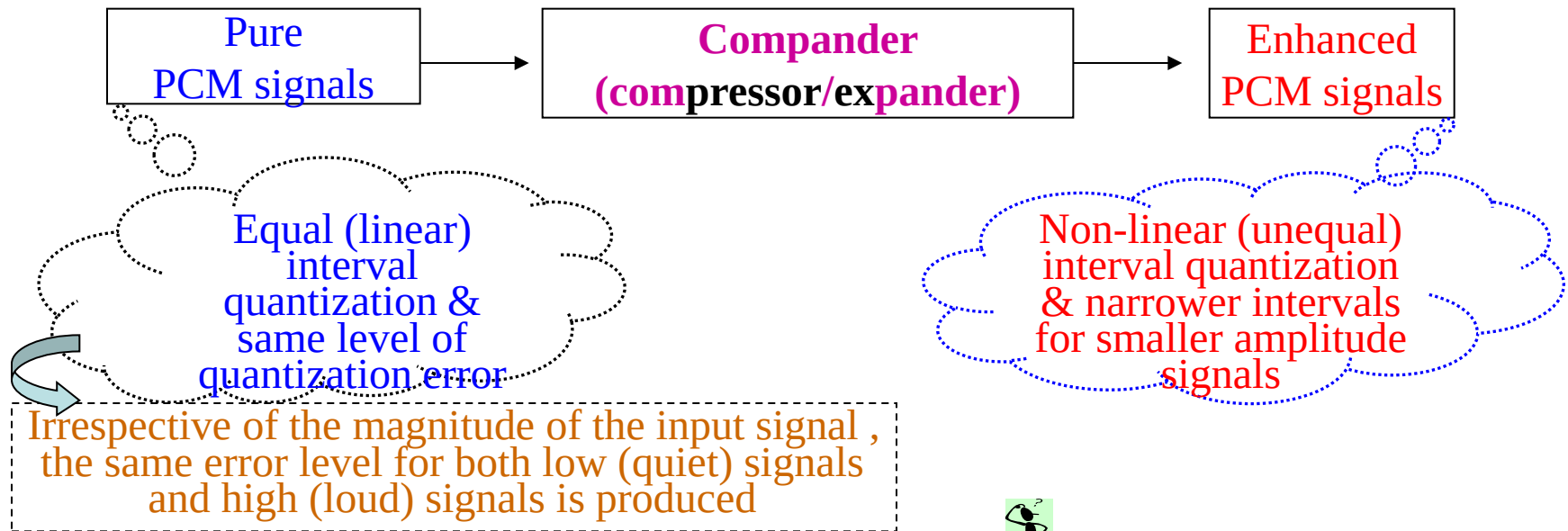


1. **Human speech: 50Hz - 10KHz** (4KHz in a plain-old-telephone system)
 - $2 \times 10K$ or $2 \times 8K$ sps $\Leftarrow$ monaural (mono) speech
 - $(2 \times 10K) \times 2$ or $(2 \times 8K) \times 2$ sps $\Leftarrow$ **stereophonic** speech
 - ideally, 12 bits/sample
2. **Human audible music: 15Hz - 20KHz**
 - $2 \times 20K$ sps $\Leftarrow$ monaural (mono) music
 - $(2 \times 20K) \times 2$ sps $\Leftarrow$ **stereophonic** music
 - ideally, 16 bits/sample

sps: samples per sec.

PCM Speech(1)

- Human Voice over PSTN
 - 200Hz-3.4Khz bandlimiting channel: about less than 4Khz
 - 8K(2×4K) sps, 8bits/sample : ITU-T G.711(PCM) recommendation
 - Companding (compressing/expanding)
 - 1-bit: polarity, 3-bit: segment code, 4-bit: quantization code



Why companding ?

Because the human ears are more sensitive to noise on quiet signals than it is on loud signals. Hence the effect of quantization noise (error) can be reduced with companding

- **Companding Example:** 5-bit per sample (1-bit polarity, 2-bit segment code, & 2-bit quantization code)



PCM Speech(4)

- Two **Compressing Codewords** for PCM
 - μ -law: North America & East Asia
 - A-law: Europe

"Signed & magnitude" representation

	μ -law	A-law
+127	1 0000000	1 1111111
+96	1 0011111	1 1100000
+64	1 0111111	1 1000000
+32	1 1011111	1 0100000
+0	1 1111111	1 0000000
-0	0 1111111	0 0000000
-32	0 1011111	0 0100000
-64	0 0111111	0 1000000
-96	0 0011111	0 1100000
-127	0 0000000	0 1111111

Sign bit (polarity) 1's complement

CD-Quality Audio

Standard associated with these devices is CD-DA(Digital Audio) standard

- Human audible bandwidth: 15Hz-20Khz $\Rightarrow$ 40Ksps
- In CD-ROMs, more higher, say, 44.1Ksps(Sampled at $23\mu\text{s}$ intervals) & 16-bit/sample (65536 equal quantization intervals) used

- bit rate for channel = sampling rate $\times$ bits per sample
= $44.1 \times 10^3 \times 16 = 705.6 \text{ Kbps}$

- total rate required for stereophonic music
= $2 \times 705.6 = 1.411 \text{ Mbps}$

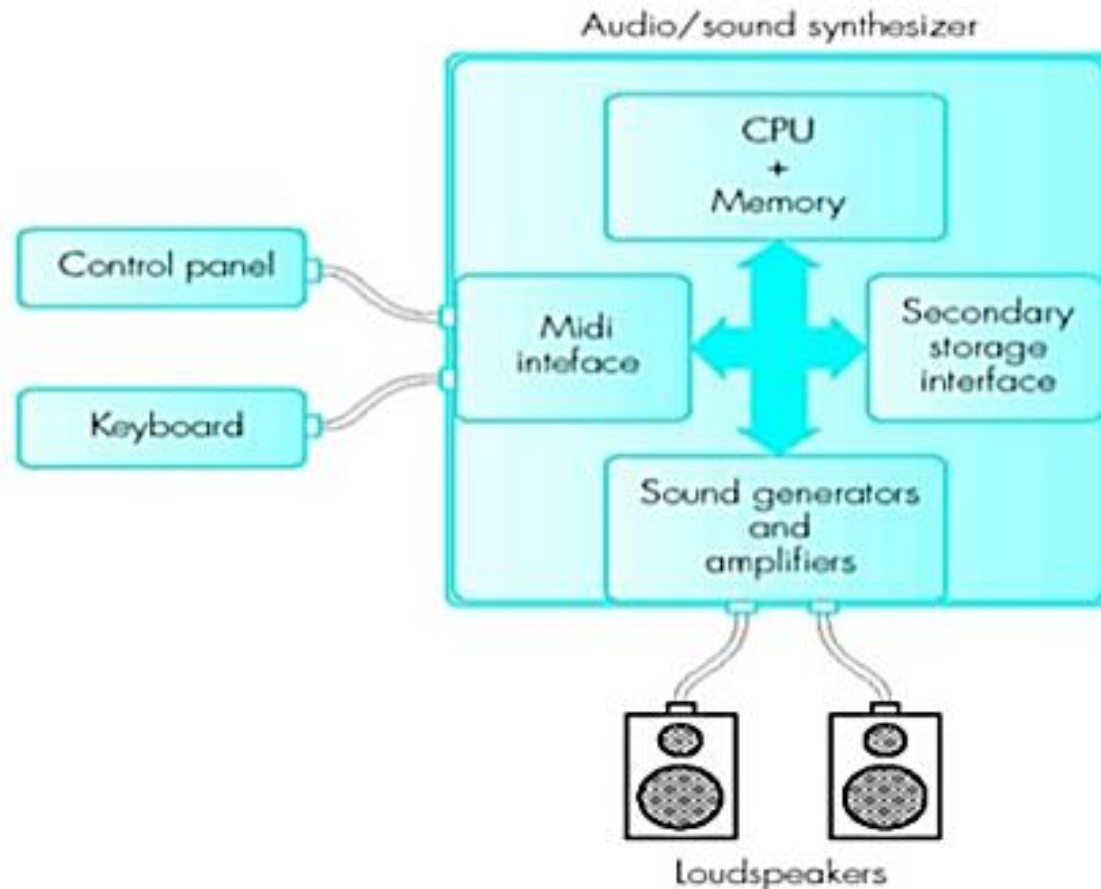
- storage capacity for a 1 hour CD-ROM title
= $1.411 \times 60 \times 60 = 634.95 \text{ Mbytes}$

this takes $(634.95 \times 10^6 \times 8)/(10 \times 10^6) = 8.5 \text{ min.}$ downloading time via a 10Mbps link network !

- Assume the CD-DA standard is being used, derive:
 - i. the time to transmit a 30 second portion of the title using a transmission channel of bit rate:
 - 64 kbps
 - 1.5Mbps

- One 30 second portion of the title = 1.411×30
= **42.33 Mbits.**
- Hence time to transmit this data:
- At 64 Kbps = $(42.33 \times 10^6) / (64 \times 10^3) =$ **661.4s**
- At 1.5 Mbps = $(42.33 \times 10^6) / (1.5 \times 10^6) =$ **28.22s**

Synthesized audio



- The computer takes input commands from the keyboard and outputs these to the sound generators which produce the corresponding sound waveform to drive the speakers

Synthesized Audio

- Synthesized audio is often used since the amount of memory required can be between *two or three orders* of magnitude less than that required to store the equivalent digitised waveform version
- The three main components of an audio synthesizer are the *computer* (with various application program), the *keyboard* (based on that of a piano) and the set of *sound generators*
- The computer takes the commands and outputs these to the sound generators which in turn produce the corresponding sound waveform – via DACs to drive the speakers

Synthesized Audio

- Pressing a key has similar effects to pressing a keyboard of a computer. For each key press a different *codeword* (**message** – indicating the *key pressed* and the *pressure applied*) is generated
- The **control panel** contains range of different *switches* and *sliders* that collectively allow the user to indicate to the program information such as the *volume* of the generated output and *selected sound effects* to be associated with each key
- To discriminate between the inputs from different possible sources a standard known set of messages (also includes the type of *connectors, cables, electrical signals, etc*) have been defined: ***Music Instrument Digital Interface*** (MIDI)

MIDI

- *Status byte* - This defines the particular event that has caused the message to be generated
- *Data bytes* – Which collectively define a set of parameters (pressure applied, identity of the key) associated with the event
- *Event*– A key being pressed
- It is important to identify the different types of instruments that generated the events
- Each instrument has a MIDI code associated with it – e.g *Piano has a code of 0 and violin 40*
- Since the music is in the form of MIDI messages it is vital to have a sound card in the client computer to interpret the sequence

2.6 Video (Motion): Broadcast TV

Video Applications

- **Entertainment**: Broadcast TV, VCR/DVD Recordings
- **Interpersonal**: Video Telephony & Videoconferencing
- **Interactive**: Video Clips on PC Windows

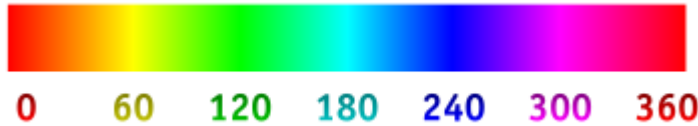
- Scanning Sequences: **Interlaced Scanning**
 - To minimize the amount of tx bandwidth, a **frame** is divided into two halves called **fields**
e.g) 525-line 50-time frame refresh rate/sec.
 - 262.5 odd lines 50-time field rate/sec.
 - 262.5 even lines 50-time field rate/sec.
- ⇒ In reality,
525-line 25-time frame refresh rate/sec.

Principles of colour TV transmission

- Colour TV transmission involves the simultaneous transmission of the **luminance** and **chrominance** components of a colour picture, with luminance predominant over chrominance
- As for the chrominance component, it is first ‘purified’ by removing the luminance component from each primary colour, resulting in what is known as *colour difference signals*:

R-Y G-Y B- Y

Broadcast TV(2)



Luminance
Brightness
Hue (Tint)
Saturation
Chrominance

■ Color Signals

■ Three **properties** of a color

- Brightness, Hue (Tint) & Saturation
- Yellow = R +G, Magenta = R+B, White=R+G+B

■ **Color production**: an equation of R, G, and B phosphors

- $0.299 R + 0.587 G + 0.114 B$ where, $0.299+0.587+0.114=1$

■ Luminance refers to the brightness of a source, the hue & the saturation called, chrominance characteristics

- say, luminance $Y_s = 0.299 R_s + 0.587 G_s + 0.114 B_s$

Y_s : magnitude of the luminance signal

R_s, G_s, B_s : magnitudes of three major colors

■ Since the Luminance signal $Y = R+G+B$ we get Two color difference signals: Blue chrominance C_b and Red chrominance C_r

- $C_b = B_s - Y_s$, $C_r = R_s - Y_s$

Broadcast TV(3)

- Chrominance Components

- Composite Video Signal for Transmission

- Y_s , C_b , and C_r signals are combined together and signal differences are scaled down before transmission

- In PAL(phase Alternating line)(color encoding system)

- $Y = 0.299 R + 0.587 G + 0.114 B$

- $U(C_b) = 0.493(B-Y) = -0.147R-0.289G+0.437B$

- $V(C_r) = 0.877(R-Y) = 0.615R-0.515G-0.1B$

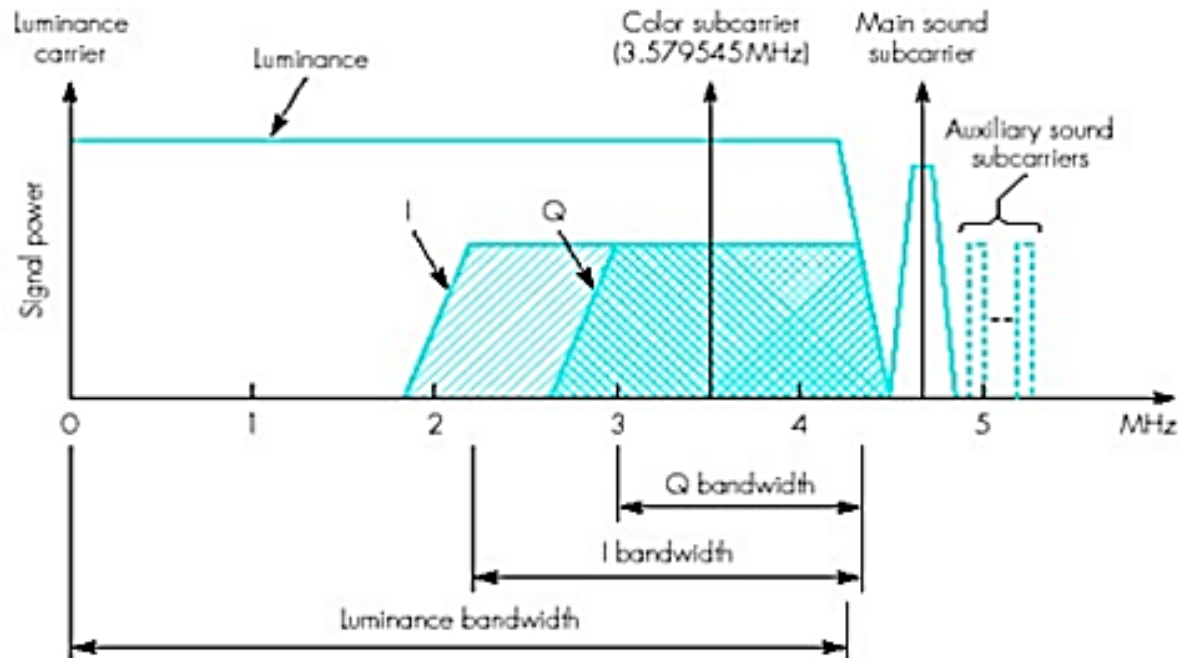
- In NTSC(National Television system committee)

- $Y = 0.299 R + 0.587 G + 0.114 B$

- $I(C_b) = 0.74(R-Y)-0.27(B-Y) = 0.599R-0.276G-0.324B$

- $Q(C_r) = 0.48(R-Y)+0.41(B-Y) = 0.212R-0.528+0.311B$

Signal Bandwidth Baseband spectrum of colour TV in NTSC System

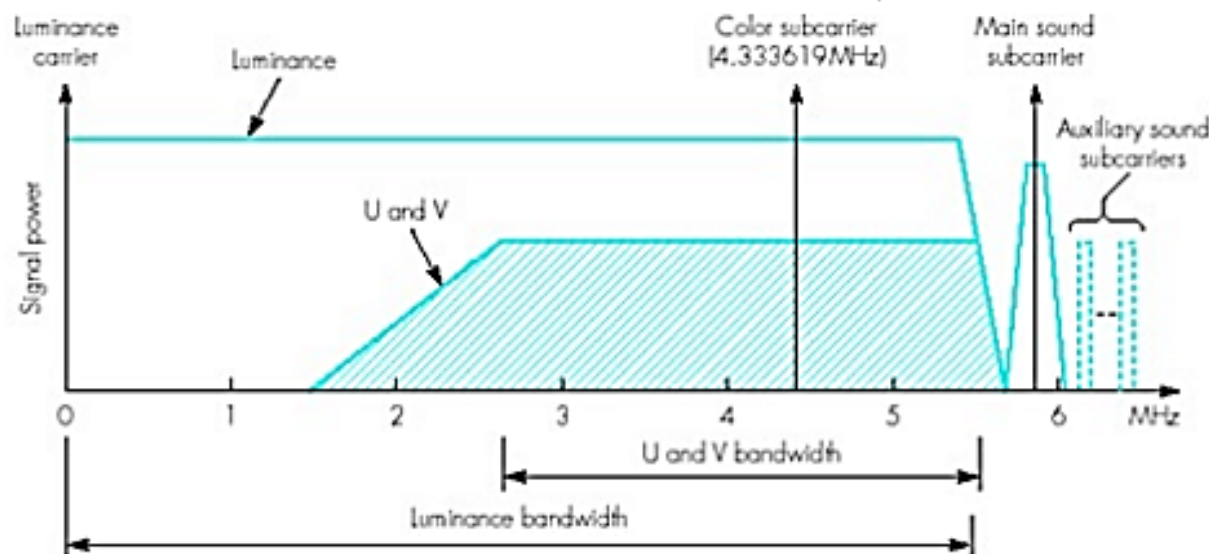


- I signal bandwidth – 2 MHz

- Q signal – bandwidth – 1 MHz

- In NTSC the eye is more responsive to the *I* signal than the *Q* signal, hence maximizing the available bandwidth and minimizing the level of interference with the luminance signal is needed

Signal Bandwidth - Baseband spectrum of colour TV in PAL System



- In PAL, the larger luminance bandwidth allows both the U and V chrominance signals to have the same modulated bandwidth
- U and V chrominance signals have the same modulated bandwidth of 3 MHz.
- The addition of the sound and video signal is called the complex baseband signal

Digital Video

- With digital television it is more usual to digitize the three component signals separately prior to their transmission to enable editing and other operations to be readily performed
- Since the eye is less sensitive for colour than it is for luminance, a significant saving in terms of resulting bit rate can be achieved by using the luminance and two colour difference signals instead of the RGB directly
- **Digitization formats** exploit the fact that the two chrominance signals can tolerate a reduced resolution relative to that used for the luminance signal

Digital Video

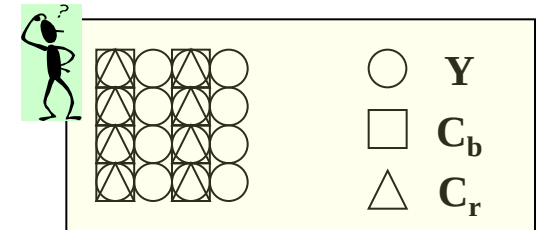
- Advantages of DV
 - Easy to store in computer
 - Easy to edit and integrate with other types
 - Easy to digitize three RGB component signals
- Transmission bandwidth is achieved by using the **luminance and two color difference signals**, instead of the RGB signals directly.
 - CCIR-601 Recommendations: standard for the digitization of video pictures

Digital Video(2)

■ 4:2:2 format(CCIR-601)

- Recommendation for use in TV studio
- Three component (analog) video signals may have bandwidths
 - up to 6Mhz for the luminance $\Rightarrow$ 12Mhz sps
 - less than 3Mhz for the two chrominance signals $\Rightarrow$ 6 Mhz sps
 - In reality, 13.5M sps for luminance, 6.75 M sps for the two chrominance signals
- In NTSC(525-line) system, total line sweep time 63.56 μ sec =
retrace time 11.56 μ sec + an active line sweep time 52 μ sec
- In PAL(625-line) system, total line sweep time 64 μ sec =
retrace time 12 μ sec + an active line sweep time 52 μ sec

Orthogonal sampling



Line sampling rate:
 $52 \times 10^{-6} \times 13.5 \times 10^6 = 702$ samples/line
 In reality, 720 samples/line

Line sampling rate:
 $52 \times 10^{-6} \times 6.75 \times 10^6 = 351$ samples/line
 In reality, 360 samples/line

4Y samples for every 2C_b and 2C_r samples(4:2:2)

Digital Video(3)

■ 4:2:2 Format Bit Rate & Storage (NTSC 525-line) *PAL 625-line*

■ The number of active (visible) lines: 480 *576*

■ The number of samples per line: 720 *720*

⇒ Resolution of luminance $Y = \underline{720 \times 480}$ *720×576*

Two chrominance signals $C_b = C_r = \underline{360 \times 480}$ *360×576*

■ Line sampling rate: 13.5sps for Y & 6.75sps for both C_b & C_r

■ Bits per sample: 8 bits

⇒ Bit rate per line = $13.5 \times 10^6 \times 8 + 2 \times (6.75 \times 10^6 \times 8) = 216\text{Mbps}$

⇒ Bits per line = $720 \times 8 + 2 \times (360 \times 8) = 11.52\text{Kbits}$

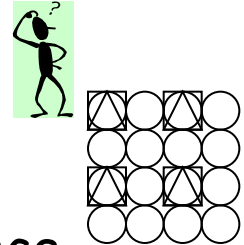
⇒ Bits per frame = 480×11.52 *576×11.52* = 5.5296Mbits *6.63555Mbits*

⇒ Bits for 1.5 hrs Video assuming 60 refresh rate = $5.5296 \times 60 \times 1.5 \times 3600$
= 223.9488GBytes *6.63555×50*

Digital Video(4)

■ 4:2:0 Format

- used in Digital Broadcast Applications
- interlaced scanning with the absence of chrominance samples in alternative lines
- 525-line system
 - $Y = 720 \times 480$ (the same as 4:2:2 format), $C_b = C_r = 360 \times 240$
- 625-line system
 - $Y = 720 \times 576$, $C_b = C_r = 360 \times 288$
- bit rate per line: $13.5 \times 10^6 \times 8 + 2 \times (3.375 \times 10^6 \times 8) = 162 \text{ Mbps}$



■ HDTV Format

- used in High-Definition Television (four times bit rate)
- 4/3 1440×1152 pixels(50/60 Hz refresh rate) & 16/9 wide-screen 1920×1152 pixels(25/30 Hz) with # of visible lines per frame 1080

Digital Video(5)

- **SIF (Source Intermediate Format), 4:1:1 Format**

- used in Video Cassette Recorders (VCRs)
- progressive (non-interlaced) scanning since it is intended for storage applications
- Half of 4:2:0 format: "Subsampling & Temporal Resolution"

- 525-line system

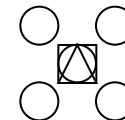
- $Y = 360 \times 240, C_b = C_r = 180 \times 120$

- 625-line system

- $Y = 360 \times 288, C_b = C_r = 180 \times 144$

- bit rate per line

- $6.75 \times 10^6 \times 8 + 2 \times (1.6875 \times 10^6 \times 8) = 81 \text{Mbps}$



Digital Video(6)

- **CIF (Common Intermediate Format), 4:1:1 format**
 - used in Video Conferencing applications
 - spatial resolution of the SIF 625-line system plus temporal resolution of the SIF 525-line system
 - $Y = 360 \times 288, C_b = C_r = 180 \times 144$
 - refresh rate: 30 Hz
 - bit rate per line: $6.75 \times 10^6 \times 8 + 2 \times (1.6875 \times 10^6 \times 8) = 81 \text{ Mbps}$
 - many variants for videoconferencing using desktop PCs or ISDN/PSTN
 - say, typically 4 or 16 64Kbps channels used
 - **4CIF**: $Y = 720 \times 576, C_b = C_r = 360 \times 288$
 - **16CIF**: $Y = 1440 \times 1152, C_b = C_r = 720 \times 576$

Digital Video(7)

- **QCIF (Quarter CIF), 4:1:1 Format**

- used in Video Telephony applications
- half spatial resolution of the CIF and either half or quarter temporal resolution of the CIF

- $Y = 180 \times 144, C_b = C_r = 90 \times 72$

- refresh rate: 15 or 7.5 Hz

- bit rate per line:

- $$3.375 \times 10^6 \times 8 + 2 \times (0.84375 \times 10^6 \times 8) = 81 \text{ Mbps}$$

- a lower version is typically used for single 64Kbps channel ISDN or PSTN with modems: **sub-QCIF(SQCIF)**

- $Y = 128 \times 96, C_b = C_r = 64 \times 48$

Digital Video(8)

■ PC Video Digitization

Digitization Format	System	Spatial Resolution	Temporal Resolution
4:2:2	525-line 625-line	Y = 640×480, C _b = C _r = 320×240 Y = 768×576, C _b = C _r = 384×288	60Hz 50Hz
SIF	525-line 625-line	Y = 320×240, C _b = C _r = 160×240 Y = 384×288, C _b = C _r = 192×144	30Hz 25Hz
CIF		Y = 384×288, C _b = C _r = 192×144	30Hz
QCIF		Y = 192×144, C _b = C _r = 96×72	15/7.5Hz
- Video capture board or S/W required - All PC monitors use “progressive (non-interlaced) scanning”			